

Namrata Paul

Nagaon, Assam, India

9864743985

namratapaul2602@gmail.com

linkedin.com/in/namrata-paul/

github.com/namratapaul26

Professional Experience

Research & Strategy Intern

Feb 2026 - May 2026

Vedantu Innovations Pvt. Ltd.

- Developed a **real-time automated YouTube analytics dashboard** with **AI-powered insights**, reducing manual reporting effort by **60%** and improving content performance tracking efficiency.
- Designed interactive **data visualization dashboards** using historical YouTube analytics data from **500+ videos**, enabling faster trend analysis and strategic decision-making.
- Integrated **Google Sheets** with **YouTube Cloud Console APIs** to automate KPI tracking and reporting workflows, improving data accuracy by **40%**.
- Collaborated cross-functionally with **research, analytics, and strategy teams**, contributing to data-driven campaign planning and operational optimization.

Web Developer Intern

Sep 2025 - Dec 2025

AlignBox

- Developed scalable and reusable **frontend components** using **React.js, HTML, CSS, and JavaScript**, improving UI consistency and reducing development time by **25%**.
- Built and integrated backend services and **REST APIs** using **Node.js and Express.js**, improving frontend-backend communication efficiency by **35%**.
- Optimized **database schema design** and application architecture, enhancing application scalability and improving system performance by **30%**.
- Collaborated with **cross-functional teams** in agile environments to support product development, debugging, and deployment workflows across multiple features.

Projects

Zomato Delivery Analytics & Intelligence Platform| *Product Analytics, PostgreSQL Warehousing, PowerBI*

May 2026 – Current

- Built an end-to-end analytics platform by integrating **3+ real-world datasets** with synthetic behavioral pipelines, processing **100K+ transactional and event records** using automated **Python ETL workflows, PostgreSQL warehousing**, and scalable data modeling techniques.
- Designed advanced **SQL analytical models** for **cohort analysis, churn prediction, delivery SLA tracking**, cancellation intelligence, and customer retention analytics, enabling actionable business insights across **20+ KPIs**.
- Developed interactive **Power BI dashboards** and scalable **FastAPI services** for real-time operational monitoring, trend forecasting, and restaurant performance analytics, improving analytical accessibility and decision visibility by **45%**.
- Tech:** Python, Pandas, PostgreSQL, SQL, Power BI, FastAPI, REST APIs, Data Warehousing, Forecasting Analytics, Faker

Car Defect Detection Model| *Deep Learning, Computer Vision*

Feb 2025 - Apr 2025

- Developed** an end-to-end car defect detection system using **YOLOv5, Faster R-CNN**, and **DETR**, trained on the **COCO dataset**, achieving **92% detection accuracy** for identifying scratches, dents, and damages.
- Enhanced localization precision through data augmentation and evaluation metrics like **mAP (0.86)** and **IoU**, improving detection reliability for autonomous vehicle inspections.
- Optimized **real-time inference pipelines**, reducing latency by **20%**, and balanced **speed-accuracy trade-offs** to ensure deployment efficiency.
- Tech:** YOLOv5, Faster R-CNN, DETR, COCO, PyTorch, TensorFlow, OpenCV, mAP, IoU, data augmentation, real-time inference

E-Commerce Product Recommender| *Machine Learning, Data Preprocessing*

Aug 2024 - Nov 2024

- Built** a product recommendation engine using **collaborative filtering, content-based filtering**, and **hybrid models**, increasing recommendation **accuracy by 23%** and improving click-through rates by **15%**.
- Streamlined data preprocessing and real-time interaction analysis**, enabling **scalable recommendations** for platforms with over **10,000 products**.
- Evaluated models using **precision (0.87)**, **recall (0.82)**, and **F1-score (0.84)** to ensure performance in large-scale e-commerce settings.
- Tech:** Python, Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, Collaborative Filtering, Content-Based Filtering, Hybrid Models, Precision-Recall Metrics

Certificates

- Supervised Machine Learning: Regression and Classification** – Stanford University Online | [\[Link\]](#) March 2025
- Cloud Computing MOOC Certification** – NPTEL Swayam (IIT Kharagpur) | [\[Link\]](#) October 2024

Technical Skills

Programming & Databases: Python, C, C++, JavaScript, SQL, MongoDB, MySQL, PostgreSQL

Machine Learning & AI: Supervised Learning, Deep Learning, Computer Vision, Classification, Regression, OCR, YOLOv5, CNNs, Data Augmentation

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, React.js, Node.js, Express.js, REST APIs

Analytics & Evaluation: EDA, Hypothesis Testing, Correlation, Probability, Dashboarding, KPI Tracking, Precision, Recall, F1-Score, mAP, IoU

Tools & Platforms: Git, GitHub, Figma, Excel, Power BI, Google Sheets

Coursework: Low-level Design (LLD), Operating Systems (OS), Data Structures, Networking, DBMS, Software Engineering

Awards & Publications

- Best Paper Award** – ICIRDST-2025 for the research paper “IoT-Enabled Deep Learning Framework for Real-Time Defect Detection in Industrial Quality Control”
- Published a peer-reviewed research paper, “Defect Detection System Using High-Resolution Cameras for Scratches/Dents” in **Springer LNNS (ICAAIML-2025)** on deep learning and computer vision.

Education

Lovely Professional University

Computer Science and Engineering

2022 – Present

Jalandhar, Punjab

Career Point Gurukul School

12th with Science — Percentage: 89.60%

2019 – 2021

Kota, Rajasthan

Christ Jyoti School

10th — Percentage: 92%

2017 – 2019

Nagaon, Assam